

PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 8 Exam Notes: Video Segmentation

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1. Temporal Video Segmentation

Video segmentation partitions a video sequence into structurally meaningful units. Temporal segmentation focuses on dividing the video stream into continuous camera sequences called **shots**.

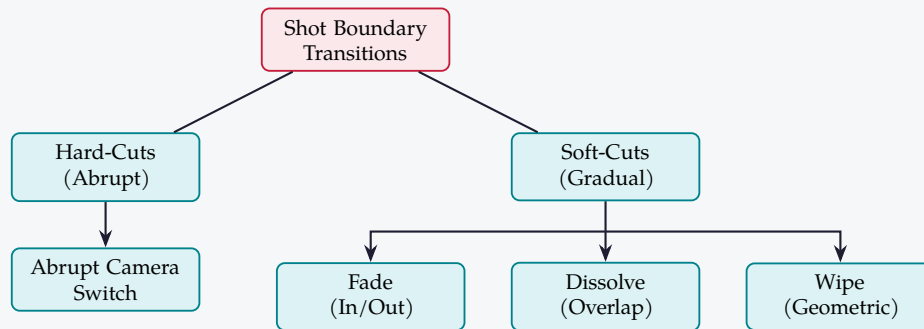
Video Shot Definitions

Video Shot: A continuous sequence of frames captured by a single camera without interruption.

Shot Boundary: The transition point between two consecutive shots. These are classified into:

- **Hard-Cuts (Abrupt):** Instantaneous transition occurring between two adjacent frames (frame t is from camera A, frame $t + 1$ is from camera B).
- **Soft-Cuts (Gradual):** Dispersed transitions taking place over multiple frames (e.g., fades, dissolves, wipes).

Shot Boundary Transition Tree



1.1 Detection Algorithms

1. **Pixel-Level Differencing:** Measures the average intensity change between pixels:

$$D(t, t-1) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} |f_t(x, y) - f_{t-1}(x, y)| \quad (1)$$

Highly sensitive to object and camera motion.

2. **Histogram-Level Differencing:** Compares the intensity histograms of adjacent frames:

$$D_H(t, t-1) = \sum_{k=0}^{L-1} |h_t(k) - h_{t-1}(k)| \quad (2)$$

Robust against minor object movements and camera pans.

3. **Edge Change Ratio (ECR):** Measures the entry and exit of edge pixels between frames, highly effective at detecting soft-cuts.

2. Spatial/Motion-Based Segmentation

Spatial segmentation separates moving foreground objects from static backgrounds.

- **Background Subtraction:** Thresholds the difference between the current frame $f_t(x, y)$ and a reference background model $B(x, y)$:

$$|f_t(x, y) - B(x, y)| > T \quad (3)$$

The background $B(x, y)$ is updated recursively (e.g., using Mixture of Gaussians).

- **Temporal Differencing:** Detects motion by subtracting adjacent frames:

$$|f_t(x, y) - f_{t-1}(x, y)| > T \quad (4)$$

More adaptive to dynamic backgrounds, but fails to segment the interior of uniformly colored moving objects (the "ghosting" effect).

- **Optical Flow:** Computes the 2D velocity vector field representing the apparent motion of pixels. Highly complex, but provides detailed motion direction.

3. Video Object Detection and Tracking

Object tracking monitors the position and trajectory of a segmented object over time.

1. **Kalman Filter:** A recursive estimator that predicts the state (position, velocity) of a moving object and corrects the prediction using noisy measurements. Operates in two steps:
 - *Predict:* Project the state ahead: $\hat{\mathbf{x}}_k^- = \mathbf{A}\hat{\mathbf{x}}_{k-1}$.
 - *Correct:* Update the estimate using the measurement: $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^-)$.
2. **Mean-Shift Tracking:** A non-parametric iterative algorithm that locates the local maxima (peak) of a probability distribution (e.g., color histogram). It moves a tracking window towards the center of gravity of the target color distribution in each frame.
3. **Particle Filters:** Uses Monte Carlo simulations (representing the target state probability distribution by a set of weighted particles) to track objects in non-Gaussian, non-linear environments.

4. Worked Examples

Temporal Differencing and Boundary Detection

Problem: Given two simplified 3×3 matrices representing consecutive video frames F_{t-1} and F_t :

$$F_{t-1} = \begin{bmatrix} 50 & 52 & 50 \\ 48 & 50 & 52 \\ 50 & 48 & 50 \end{bmatrix}, \quad F_t = \begin{bmatrix} 70 & 75 & 70 \\ 68 & 72 & 75 \\ 70 & 68 & 70 \end{bmatrix}$$

- (a) Calculate the pixel-level average difference metric $D(t, t-1)$.
- (b) Given a threshold $T_{\text{cut}} = 15$, determine if a shot boundary transition is detected between these two frames.

Solution:

(a) Average Difference Calculation ($D(t, t-1)$): Subtract the matrices element-by-element and take the absolute values:

$$\begin{aligned} |F_t - F_{t-1}| &= \begin{bmatrix} |70 - 50| & |75 - 52| & |70 - 50| \\ |68 - 48| & |72 - 50| & |75 - 52| \\ |70 - 50| & |68 - 48| & |70 - 50| \end{bmatrix} \\ &= \begin{bmatrix} 20 & 23 & 20 \\ 20 & 22 & 23 \\ 20 & 20 & 20 \end{bmatrix} \end{aligned}$$

Now, compute the average over the 9 pixels ($MN = 9$):

$$D(t, t-1) = \frac{20 + 23 + 20 + 20 + 22 + 23 + 20 + 20 + 20}{9} = \frac{188}{9} \approx 20.89$$

(b) Shot Boundary Decision:

- The computed average difference metric $D(t, t-1) = 20.89$.
- The boundary threshold $T_{\text{cut}} = 15$.
- Since $D(t, t-1) = 20.89 > T_{\text{cut}} = 15$, a shot boundary transition is indeed detected between frame $t-1$ and frame t .

Quick Revision — Module VIII

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Key Formulas at a Glance

Pixel Difference Metric	$D(t, t-1) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f_t(x, y) - f_{t-1}(x, y) $
Histogram Difference	$D_H(t, t-1) = \sum_{k=0}^{L-1} h_t(k) - h_{t-1}(k) $
Background Subtraction	$ f_t(x, y) - B(x, y) > T$
Temporal Differencing	$ f_t(x, y) - f_{t-1}(x, y) > T$
Kalman State Prediction	$\hat{\mathbf{x}}_k^- = \mathbf{A}\hat{\mathbf{x}}_{k-1}$
Kalman Measurement Update	$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^-)$

One-Line Definitions

- **Hard-Cut:** An abrupt change between two consecutive frames due to camera switching, marking a shot boundary.
- **Soft-Cut:** A gradual, multi-frame transition (such as fade, dissolve, or wipe) between two video shots.
- **Background Subtraction:** A spatial differencing technique that isolates foreground objects by subtracting a background model.
- **Temporal Differencing:** Isolating motion by thresholding the difference between consecutive video frames.
- **Mean-Shift:** An iterative peak-search algorithm that tracks target centers by maximizing color probability distributions.
- **Optical Flow:** A vector field representing the apparent velocity of moving pixels across consecutive frames.

Frequently Asked / Viva Questions

1. What is the difference between a hard-cut and a soft-cut transition?

A hard-cut is an abrupt, instantaneous shot transition occurring between two adjacent frames. A soft-cut is a gradual transition (such as a fade, dissolve, or wipe) that spans multiple frames.

2. Why is histogram-level differencing preferred over pixel-level differencing for shot boundary detection?

Pixel-level differencing is highly sensitive to camera panning, zoom, and local object motion. Histogram-level differencing is robust to these changes since the overall color distribution remains similar.

3. Explain the ghosting effect in temporal differencing.

Frequently Asked / Viva Questions (contd.)

If a moving object has a uniform interior color, subtracting adjacent frames yields differences only at the leading and trailing edges, leaving the interior segmented as background (hollow "ghost" region).

4. How does Background Subtraction model dynamic elements like swaying trees?

By recursively updating the background model using a mixture of Gaussians (MoG) that models multiple intensity distributions for each pixel location.

5. State the two phases of the Kalman Filter.

(1) The Prediction phase (projects the current state and error covariance estimate forward in time), and (2) the Correction/Measurement update phase (adjusts the estimate using the new sensor measurement).

6. What is the role of the Kalman Gain K_k ?

The Kalman gain decides the weighting between the predicted state estimate and the new measurement. If measurement noise is low, K_k weights the measurement heavily; if high, it weights the prediction.

7. Explain the difference between object detection and object tracking.

Object detection identifies and locates objects of interest in individual frames. Object tracking links these detections across consecutive frames to trace the object's path.

8. How does the Mean-Shift algorithm track a target?

It iteratively computes the mean shift vector (displacement between the current window center and the color distribution's center of gravity) and shifts the window until convergence.

9. What are the types of gradual transitions (soft-cuts) in video?

(1) Fade-in/out (gradual change from/to a solid black frame), (2) Dissolve (one scene overlaps and replaces another gradually), and (3) Wipe (a geometric line sweeps across the screen).

10. What is the limitation of optical flow estimation?

Optical flow is computationally extremely intensive, making it difficult to implement in real-time without specialized hardware, and it is sensitive to illumination changes.

Hard-Cut vs. Soft-Cut Shot Boundaries

Attribute	Hard-Cut Transition	Soft-Cut (Gradual) Transition
Duration	Exactly 1 frame boundary	Spans multiple frames (typically 10-30)
Visual transition	Abrupt camera scene switch	Smooth fade, overlap (dissolve), or sweep (wipe)
Detection Metric	High spike in frame differences	Broad, low-amplitude peak in differences